

Strong completeness of two dyadic floor sequences

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13 September 2026

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Abstract

For positive real numbers α and β with irrational ratio, we prove that the nonzero values of the two sequences $\lfloor 2^n \alpha \rfloor$ and $\lfloor 2^n \beta \rfloor$ form a strongly complete set: every finite deletion leaves a set whose distinct-element sums contain all sufficiently large integers. We construct finite integer meshes from an explicit coefficient certificate and propagate their gap bounds through every subsequent layer. Sufficiently long intervals between binary events then produce permanent descent of a modular gap invariant. Conversely, finite-event decay and a digit-budget estimate yield long rational-approximation windows with few events. Compactness of ratios of sparse binary sums forces a large event cost between exact layers, contradicting the spacing bound obtained from permanent descent. These arguments establish part (i) of Erdős Problem 354 and the stated finite-deletion strengthening.

Theorem and scope

For $\alpha, \beta > 0$, set

$$A_{\alpha, \beta} = \{\lfloor 2^n \alpha \rfloor, \lfloor 2^n \beta \rfloor : n \in \mathbb{N}\} \setminus \{0\}.$$

Theorem. If α/β is irrational, then for every finite set $F \subseteq \mathbb{Z}$, there exists an integer H such that every integer $m \geq H$ is a sum of distinct elements of $A_{\alpha, \beta} \setminus F$.

In particular, every sufficiently large integer is a finite-index sum of the interleaving of the two floor sequences. In that indexed formulation, each index is used at most once; different indices may have the same value. The theorem above gives the stronger conclusion formulated in terms of distinct values.

The [problem statement](#) attributes the dyadic completeness question to Graham and records Hegyvári's earlier result when one parameter is dyadic rational and the other is not. Hegyvári's broader conjecture concerns ratios other than integral powers of two, with at least one non-dyadic parameter. Our theorem treats the irrational-ratio case, with base exactly 2; the rational-ratio cases of that broader conjecture and the variable-base question in part (ii) are different statements.

We give the argument in full below, including the finite certificate in Appendix A. Our [Lean formalization](#) proves the indexed and strong-completeness targets under exactly the hypotheses above. This exposition follows its use of good rational approximants, an explicit decay potential,

and a $T^{3/4}$ bound in the final counting argument. The original English manuscript is retained as the fixed pre-formalization source. Our contribution statement describes the use of Chatgpt-6 Astra in both the mathematical development and the Lean formalization.

1. Normalization, indices, and two different notions of gap

Start with the original parameters $\alpha_0, \beta_0 > 0$. Choose an integer k such that $1 < 2^k \alpha_0 / \beta_0 < 2$. Equality at an endpoint would make the original ratio a rational power of two, so cannot occur. Choose nonnegative integers u, v with $u - v = k$. Replacing the two parameters by $\alpha_1 = 2^u \alpha_0$ and $\beta_1 = 2^v \beta_0$ deletes only finite prefixes. A further common multiplication by a sufficiently large power 2^T gives

$$N = \lfloor \beta \rfloor < M = \lfloor \alpha \rfloor < 2N, \quad N \geq 2.$$

Indeed, both $\alpha_1 - \beta_1$ and $2\beta_1 - \alpha_1$ are positive; after multiplication they dominate the bounded floor errors. No downward or arbitrary real scaling is used. A complete retained tail is enough to prove completeness of the original set.

Write

$$a_i = \lfloor 2^i \alpha \rfloor, \quad b_i = \lfloor 2^i \beta \rfloor, \quad a_{i+1} = 2a_i + u_i, \quad b_{i+1} = 2b_i + v_i,$$

where $u_i, v_i \in \{0, 1\}$. The inequalities

$$b_i < a_i < 2b_i \leq b_{i+1}$$

persist. Thus the actual sorted order is $b_i, a_i, b_{i+1}, a_{i+1}, \dots$, with each next weight at most twice its predecessor. In particular the retained sequences have no repeated values between them.

Events are indexed by the arrival layer:

$$\mathcal{T} = \{t \geq 1 : (u_{t-1}, v_{t-1}) \neq (0, 0)\}, \quad K_n = |\mathcal{T} \cap [1, n]|.$$

A digit pair u_i, v_i describes departure from layer i , whereas the associated event has position $i + 1$. Define

$$P_n = P(a_i, b_i : 0 \leq i < n), \quad S_n = \sum_{i < n} (a_i + b_i),$$

$$L_n = a_n + b_n, \quad D_n = \gcd(a_n, b_n), \quad X_n = P_n \bmod D_n.$$

Here P includes the empty sum; each original index can be used at most once in each representation. P_n does not use a_n, b_n , although K_n counts the conversion that determines them.

If the event set were finite, both sequences would double exactly after some n_0 . Dividing by 2^i and taking limits would give $\alpha = a_{n_0}/2^{n_0}$ and $\beta = b_{n_0}/2^{n_0}$, contradicting the irrational ratio. Thus the event set is infinite for the theorem's parameters.

For a nonempty subset X of $\mathbb{Z}/d\mathbb{Z}$, let $h(X)$ be the number of residues in its longest consecutive missing run around the circle, and set $h(X) = 0$ when X is full. Put $h_n = h(X_n)$. For a finite integer set W with at least two elements define

$$\text{span}(W) = \max W - \min W, \quad \text{gap}(W) = \max\{w_{j+1} - w_j\},$$

where the w_j are its distinct elements in increasing order. A gap of k corresponds to at most $k - 1$ consecutive missing integers.

1.1 A fixed bound for every prefix

For sorted positive weights c_j with $c_{j+1} \leq 2c_j$,

$$c_{j+1} - \sum_{i \leq j} c_i \leq c_j - \sum_{i < j} c_i \leq c_0.$$

When adding a weight, the new subset sums are the old set and a translate. If their convex hulls are disjoint, the only new gap is bounded by the expression above; if they intersect, gaps do not increase. Hence $\text{gap}(P_n) \leq N$ on $[0, S_n]$.

We also have $S_n \geq a_n$ for $n \geq 2$. Indeed, $S_2 \geq 3(M + N)$ and $a_2 \leq 4M + 3$, whence $S_2 - a_2 \geq 3N - M - 3 \geq N - 2 \geq 0$. The induction step is

$$S_{n+1} - a_{n+1} = (S_n - a_n) + b_n - u_n \geq 0.$$

Using Lemma 2.3 and $D_n \leq b_n \leq a_n$ gives

$$0 \leq h_n \leq N - 1 \quad (n \geq 2). \quad (1.1)$$

The single-column identity $a_j - \sum_{i < j} a_i = M + \sum_{i < j} u_i > 0$ and its b analogue also imply $S_n < L_n$.

2. Three finite lemmas

Lemma 2.1: erosion by one translate

For a nonempty $X \subseteq \mathbb{Z}/d\mathbb{Z}$,

$$h(X \cup (X + 1)) = \max(0, h(X) - 1). \quad (2.1)$$

The complement is the intersection of the old complement with its one-step translate. Each missing run of length r is shortened to $\max(0, r - 1)$. Nonemptiness excludes the all-missing circle.

Lemma 2.2: propagation of a finite integer mesh

If $\text{span}(W) \geq c > 0$ and $\text{gap}(W) \leq k$, then

$$\text{gap}(W \cup (W + c)) \leq k, \quad \text{span}(W \cup (W + c)) = \text{span}(W) + c. \quad (2.2)$$

The two convex hulls intersect or touch, and their endpoints belong to the union. An empty interval between consecutive union points cannot cross a hull endpoint in its interior. It is therefore contained in one of the hulls, where its length is at most k .

Consequently, if future weights obey $c_{i+1} \leq 2c_i$ and the initial span is at least c_1 , this construction keeps the same gap bound indefinitely: adding c_i makes the span at least $2c_i \geq c_{i+1}$.

Lemma 2.3: projection to any smaller modulus

If $\text{span}(W) \geq m \geq 1$ and $\text{gap}(W) \leq k$, then

$$h(W \bmod m) \leq k - 1. \quad (2.3)$$

Translate W analytically so that $\min W = 0$; this merely rotates its residues, and does not introduce negative original summands. The case $m = 1$ is immediate. For $m \geq 2$, gaps among the points of $W \cap [0, m)$ are at most k . Let w_- be its last point and w_+ the first point of W at least m . The latter exists by the span assumption. They are consecutive in W , so $m - w_- \leq w_+ - w_- \leq k$. This also bounds the wraparound interval to zero. Adding residues from other points only reduces gaps.

We will preserve an actual integer mesh, not just a density in a changing group.

3. A long exact block and its finite coefficient certificate

At a fixed layer n , write

$$A = a_n = dp, \quad B = b_n = dq, \quad (p, q) = 1, \quad q < p < 2q,$$

and put $E = P_n$, $S = S_n < d(p + q)$, $X = E \bmod d$, $H = h(X)$, $k = \max(1, H)$. Thus $1 \leq k \leq d$.

Use ℓ exact doubling pairs at indices $n, \dots, n + \ell - 1$, with $K = 2^\ell$. The next conversion is nonzero, $(u_1, v_1) \neq (0, 0)$; the following two conversions $(u_2, v_2), (u_3, v_3)$ are arbitrary. The three new pairs are

$$\begin{aligned} dKp + u_1, & \quad dKq + v_1, \\ 2dKp + 2u_1 + u_2, & \quad 2dKq + 2v_1 + v_2, \\ 4dKp + 4u_1 + 2u_2 + u_3, & \quad 4dKq + 4v_1 + 2v_2 + v_3. \end{aligned}$$

They use indices $n + \ell, \dots, n + \ell + 2$ and belong to P_r for $r = n + \ell + 3$. Define

$$F = q(p - 1), \quad B_* = S + d(F + p + q) + 22, \quad K_* = 2q(p - 1) + 4(p + q) + 64,$$

and assume $K \geq K_*$.

3.1 The old coefficient interval

The exact block provides $d\{px + qy : 0 \leq x, y < K\}$. When $K \geq p$ this includes d times the interval

$$[F, (p + q)(K - 1) - F]. \quad (3.1)$$

For $F \leq z \leq p(K - 1)$ choose $0 \leq y < p$ with $qy \equiv z \pmod{p}$, then $x = (z - qy)/p$ belongs to $[0, K - 1]$. Reflecting the coefficients in the center of the square gives the other interval, whose left endpoint is $q(K - 1)$. They intersect because $q < p$.

3.2 Two alternative offsets whose constants differ by one

Suppose two subset sums of the six new weights have the form

$$\sigma_0 = dKl_0(p, q) + c, \quad \sigma_1 = dKl_1(p, q) + c + 1, \quad 0 \leq c < c + 1 \leq 22.$$

They are alternatives, not sums to be used together. Set

$$L = \max(l_0, l_1), \quad U = \min(l_0, l_1) + p + q, \quad J = [dKL + B_*, dKU - B_*].$$

Every $z \in J$ whose residue lies in $(X + c) \cup (X + c + 1)$ has a legal representation. Choose the corresponding offset and $f \in E$ so that $d \mid z - \sigma_j - f$. Since $0 \leq f \leq S$, the two endpoint bounds ensure

$$F \leq (z - \sigma_j - f)/d \leq (p + q)(K - 1) - F.$$

Equation (3.1) supplies the remaining old-block sum. The three groups of indices are disjoint. The bound B_* pays for actual representatives, not merely residues. By (2.1), this proves

$$\text{Every } k \text{ consecutive integers in } J \text{ include a point of } P_r. \quad (3.2)$$

3.3 The finite cone certificate

The mask table in Appendix A supplies a chain of the intervals $[L, U]$ satisfying

$$\begin{aligned} L_{\text{first}} &\leq p + 2q, & U_{\text{last}} &\geq 7p + 6q, \\ U_i - L_i &> 0, & U_i - L_{i+1} &> 0, & U_{i+1} - L_i &> 0. \end{aligned}$$

Every quantity is an integer, so each strict margin is at least one. To check a homogeneous form $ap + bq$ on $q < p < 2q$, write $p = 2x + y$, $q = x + y$ with $x, y > 0$. The form is nonnegative on the closed cone exactly when $2a + b \geq 0$ and $a + b \geq 0$; if its coefficients are not both zero, it is positive in the open cone.

The first two pairs have 16 subset masks. There are three possible nonzero first digits and four second digits, hence 12 templates. To each pair of masks with constant difference one, add the same subset of the third pair to both offsets. This shifts coefficients by 0, $4p$, $4q$, or $4p + 4q$; it does not change the constant difference. For all four third digits the constants stay between 0 and 22.

Appendix A contains all 125 nodes and 113 strict links. The standalone [checker](#) reconstructs coefficients from masks, checks the cone inequalities, checks all 500 third-digit instances, and compares the two proof appendices and the JSON. In Lean, the concrete checker is evaluated by kernel reduction, and the soundness theorem proves that acceptance implies these inequalities and legal six-weight representations for every admissible pair p, q . Thus the finite data supply a universal coefficient lemma.

4. Connecting meshes rather than complete intervals

Since S is an integer, $S \leq d(p + q) - 1$. Thus

$$\begin{aligned} dK - 2B_* &\geq d[2F + 4(p + q) + 64] - 2[d(p + q) - 1 + d(F + p + q) + 22] \\ &= 64d - 42 \geq 22d. \end{aligned} \quad (4.1)$$

This includes $d = 1$, where the last margin is exactly 22. Each actual node and each adjacent overlap has width at least $22d$.

Put

$$L_* = dK(p + 2q) + B_*, \quad U_* = dK(7p + 6q) - B_*, \quad I = [L_*, U_*].$$

The constants c can differ between nodes, so short internal gaps alone are not enough to connect them. For a node $J_i = [l_i, r_i]$, the starting points of its length- k integer windows form $J_i^- = [l_i, r_i - k + 1]$. As $k \leq d$, (4.1) ensures that these shortened intervals remain nonempty and adjacent ones still intersect. Their union is connected and covers $[L_*, U_* - k + 1]$. No monotonicity of the node endpoints is required.

Every length- k integer window in I is therefore contained in a single node, where (3.2) applies. Consequently, the actual subset-sum set $W = P_r \cap I$ satisfies

$$\begin{aligned} \min W &\leq L_* + k - 1, & \max W &\geq U_* - k + 1, & \text{gap}(W) &\leq k, \\ \text{span}(W) &\geq U_* - L_* - 2(k - 1). \end{aligned} \quad (4.2)$$

The next unused smallest weight has $b_r \leq 8dKq + 15$. Since $6p - 4q \geq 1$,

$$U_* - L_* - (8dKq + 15) \geq dK - 2B_* - 15 \geq 22d - 15.$$

Hence

$$\text{span}(W) - b_r \geq 22d - 15 - 2(k - 1) \geq 20d - 13 > 0. \quad (4.3)$$

This is a mesh of actual integers with sufficient span to propagate.

5. Permanent descent, uniform over every future continuation

Theorem 5.1

Under Section 3's hypotheses, for every legal continuation and every $t \geq r$,

$$h_t \leq \max(0, h_n - 1). \quad (5.1)$$

If $h_n \leq 1$, the whole continuation is complete.

Proof. For each $t \geq r$ define the actual set

$$W_t = W + P(a_i, b_i : r \leq i < t) \subseteq P_t.$$

Apply Lemma 2.2 in the sorted order of unused weights. Its hypotheses hold by (4.3), and each next weight is at most twice the preceding one. Thus

$$\text{gap}(W_t) \leq k, \quad \text{span}(W_t) \geq b_t \geq D_t.$$

Lemma 2.3 gives $h(W_t \bmod D_t) \leq k - 1$, and P_t contains W_t , proving (5.1). When $k = 1$, the W_t are full integer intervals with the same fixed left endpoint and unbounded right endpoints. Their union contains a half-line. $\square$

This does not assert that h_t is monotone at each individual conversion. It asserts a permanently lower bound for all future moduli after a qualifying update. Neither a majority hypothesis nor a changed endpoint ratio is required.

Since $K_* \leq 16p^2$ and $p \leq a_n < (M + 1)2^n$, the sufficient length condition

$$\ell \geq 2n + C_M, \quad C_M = 16(M + 1)^2 \quad (5.2)$$

implies $K \geq K_*$, since $2^{C_M} \geq 16(M + 1)^2$. We use this explicit, non-optimal constant in the formalization. It is independent of every future word and modulus.

6. Event spacing by well-founded descent

Suppose the arrival-event set is infinite. For consecutive events $n < m$, call the gap qualifying if $m - n \geq 2n + C_M$.

There is no nonzero conversion between arrival layers n and m . Accordingly, the $\ell = m - n$ pairs at indices $n, \dots, m - 1$ form an exact doubling block, and the first nonzero conversion produces the pair at m . The three new pairs occur at $m, m + 1, m + 2$. Theorem 5.1 therefore gives

$$h_t \leq \max(0, h_n - 1) \quad (t \geq m + 3).$$

We show that qualifying starting layers cannot be unbounded for an incomplete sequence. Otherwise, begin at any qualifying layer. If its gap invariant is at most one, Theorem 5.1 already gives completeness. If it is larger, choose the next qualifying starting layer beyond the preceding permanent update. Its invariant is strictly smaller. Well-founded descent on the natural numbers rules out indefinite repetition. Notice that this argument uses the permanent bound after each update, not monotonicity at every layer.

Thus, under incompleteness, consecutive sufficiently late events satisfy

$$m < 3n + C_M,$$

and hence $m \leq 4n$ once $n \geq C_M$. Equivalently, after increasing a fixed threshold n_0 if necessary, every integer $n \geq n_0$ has an arrival event in $(n, 4n]$. To see this, take the last event at or before n and its successor; the last event is eventually beyond the threshold for the consecutive-event bound.

7. Reduction to the finite-event contradiction

For the normalized parameters, irrationality gives infinitely many events by Section 1. If their interleaved sequence were incomplete, Section 6 would give a fixed bound on event spacing. Sections 8–11 contradict this combination of irrationality, incompleteness, and bounded spacing.

To recover the theorem for the original parameters, let F be a finite set of deleted values and choose an upper bound for $F \cup \{0\}$. We may carry out the upward dyadic shifts in Section 1 so that both retained tails lie above this bound. Their ratio is still irrational. Their values strictly interlace, so a finite-index representation from these tails is a sum of distinct values of $A_{\alpha, \beta} \setminus F$. Completeness of these tails proves strong completeness of the original set. Taking $F = \emptyset$ also gives the indexed conclusion: select one original index for each of the finitely many represented values.

8. Finite-event decay (FE), including the changing-period comparison

This section re-proves the internal estimate used below. To avoid confusing a window deficit with a gcd, write

$$B_n = L_n - S_n = M + N + \sum_{i < n} (u_i + v_i) > 0, \quad Q_n = L_n - |P_n|, \quad G_n = |P_{n+1}| - 2|P_n| \geq 0.$$

The two sets P_n and $P_n + L_n$ are disjoint. With $w_n = u_n + v_n$,

$$P_{n+1} = P_n + \{0, a_n, b_n, L_n\}, \quad Q_{n+1} = 2Q_n + w_n - G_n. \quad (8.1)$$

Periodically extend the indicator of missing positions in $[0, L_n - 1]$ to f_n . For a function of period L , put

$$J_t(f) = \sum_{x \bmod L} |f(x+t) - f(x)|.$$

Then $J_{-t} = J_t$, $J_{s+t} \leq J_s + J_t$, and J_1 counts twice the number of cyclic missing runs. Internal runs have length at most $N - 1$; the terminal padding run has length at most $B_n - 1$. Thus

$$Q_n \leq (N/2)J_1(f_n) + B_n. \quad (8.2)$$

Regard P_n as a subset of $\mathbb{Z}/L_n\mathbb{Z}$. Every residue in $(P_n + a_n) \setminus P_n$ has an actual new representative outside both basic copies. Distinct residues give distinct integers. Therefore

$$J_{a_n}(f_n) \leq 2G_n, \quad J_{b_n}(f_n) = J_{a_n}(f_n), \quad J_{2a_n}(f_n) \leq 4G_n. \quad (8.3)$$

Set $a = a_n, b = b_n, L = a + b, u = u_n, v = v_n, w = u + v$. The new period is $L' = 2L + w$. Before adding the two single translates, the missing word is $f_n f_n 1^w$; these translates fill exactly G_n holes. Comparing with the old periodic extension yields

$$\sum_{0 \leq x < L'} |f_{n+1}(x) - f_n(x)| \leq G_n + w. \quad (8.4)$$

The old function is extended using its old period. The two periods have not been identified; the final w positions are explicitly paid for.

8.1 Nonzero events control the unit boundary

Suppose $u = 1$. The new shift is $a' = 2a + 1$, whose nonwrapping segment is $0 \leq x < 2b + v$. By (8.3) at the next layer and two applications of (8.4),

$$\sum_{x=0}^{2b+v-1} |f_n(x+2a+1) - f_n(x)| \leq 2G_{n+1} + 2G_n + 2w.$$

Use its first $2b$ terms, $x = 0, \dots, 2b-1$. They are legitimate because $x+a' \leq 2L < L' = 2L+1+v$. Requiring $x+a' < 2L$ would impose an unnecessary old-period cutoff. Cancelling the $2a$ shift costs at most $4G_n$. For $\tau(x) = |f_n(x+1) - f_n(x)|$ this gives, on the old circular arc $I = [L-2b, L-1]$,

$$\sum_I \tau \leq 2G_{n+1} + 6G_n + 2w.$$

The arcs I and $I + b$ cover the old circle since $b < a < 2b$. Moreover,

$$\sum_{x \bmod L} |\tau(x+b) - \tau(x)| \leq 2J_b(f_n) \leq 4G_n.$$

It follows that

$$J_1(f_n) \leq 16G_n + 4G_{n+1} + 4w. \quad (8.5)$$

If $u = 0, v = 1$, use the new period $2L + 1$ and shift $2a$. Take the L wrapping positions $x = 2b + 1, \dots, 2b + L$, which are valid because $b < a$. The wrapped coordinate is $x - 2b - 1$, equivalent modulo the old period to $x + 2a - 1$. Equations (8.3)–(8.4) give

$$J_{2a-1}(f_n) \leq 2G_{n+1} + 2G_n + 2, \quad J_1(f_n) \leq 2G_{n+1} + 6G_n + 2,$$

which is stronger than (8.5). This covers all three nonzero digit types.

8.2 A potential for two-step decay

From (8.2) and (8.5),

$$G_n + G_{n+1} \geq \frac{Q_n - B_n - 4N}{8N}.$$

Applying (8.1) twice at a nonzero event gives

$$Q_{n+2} \leq \left(4 - \frac{1}{8N}\right) Q_n + \frac{B_n}{8N} + \frac{13}{2}.$$

Set

$$z_n = \frac{Q_n}{2^n}, \quad \rho = 1 - \frac{1}{32N}, \quad \sigma = 1 - \frac{1}{64N}.$$

Since $B_n < 3N + 2n$, we obtain

$$\begin{aligned} z_{n+1} &\leq z_n & (w_n = 0), \\ z_{n+1} &\leq z_n + 2^{-n} & \text{always}, \\ z_{n+2} &\leq \rho z_n + 2(n+1)2^{-n} & (w_n > 0). \end{aligned}$$

We absorb the errors into the potential

$$V_n = z_n + 9(n+1)2^{-n}.$$

For $N \geq 2$ we have $\rho \geq 63/64$ and $\rho \leq \sigma^2$. The elementary inequality

$$2(n+1)2^{-n} + 9(n+3)2^{-(n+2)} \leq \rho 9(n+1)2^{-n}$$

shows that $V_{n+2} \leq \rho V_n \leq \sigma^2 V_n$ at a nonzero event. At a zero event $V_{n+1} \leq V_n$, and always $z_{n+1} \leq V_n$.

Partition a finite interval of conversions into one-step zero blocks and two-step blocks beginning with a nonzero event. A two-step block contains at most two events. Since $0 < \sigma < 1$, its factor σ^2 pays for both; a zero block needs no decay factor. There can be at most one final unpaired conversion. Its estimate $z_{n+1} \leq V_n$ is absorbed by a factor 2, because $\sigma \geq 1/2$. Induction on the interval length therefore gives, for all $m, d \geq 0$,

$$z_{m+d} \leq 2V_m \sigma^{K_{m+d}-K_m}.$$

At $m = 0$, $V_0 = M + N + 8$. Using $1 - x \leq e^{-x}$, we conclude

$$Q_n \leq C_0 2^n e^{-aK_n}, \quad C_0 = 2(M + N + 10), \quad a = \frac{1}{64N}. \quad (\text{FE})$$

All cardinalities here count distinct subset-sum values. This potential argument is the form of the error estimate used in the formalization.

8.3 A contiguous-seed lower bound

Let $e_n = S_n + 1 - |P_n| \leq Q_n$, and let R_n be the largest width of an integer interval contained in P_n . Its at most $e_n + 1$ represented runs imply

$$R_n + 2 \geq \frac{S_n + 2}{e_n + 1}.$$

For $n \geq 1$, $S_n \geq (M + N)2^{n-1}$, and $2^n e^{-aK_n} \geq 1$ because $K_n \leq n$ and $a < \log 2$. Therefore

$$R_n + 2 \geq c_0 e^{aK_n}, \quad c_0 = \frac{M + N}{2(C_0 + 1)} > 0. \quad (\text{FE-R})$$

9. Digit-budget propagation (DB)

Let $\theta = \alpha/\beta \in (1, 2)$ and choose relatively prime p, q , $q \geq 2$, with $|\theta - p/q| < 1/q^2$. At prefix depth n put

$$\lambda = 2^n \beta, \quad k = \lceil \log_2(8q) \rceil, \quad K = 2^k, \quad E_{n,k} = \sum_{i=n}^{n+k-1} (\{2^i \alpha\} + \{2^i \beta\}).$$

For either column, $r_{i+1} = 2r_i - \epsilon_i$ implies $\sum_{i=n}^{n+k-1} r_i = \sum_{i=n}^{n+k-1} \epsilon_i - r_n + r_{n+k}$. Thus

$$0 \leq E_{n,k} < 2(K_{n+k} - K_n) + 2. \quad (9.1)$$

The ideal suffix sums are $\lambda(\theta x + y)$ for $0 \leq x, y < K$. Each corresponding actual sum is shifted down by an amount in $[0, E_{n,k}]$.

The q phases $j\theta \bmod 1$, $0 \leq j < q$, lie within $1/q$ of a uniformly spaced q -grid, so their maximum circular gap is less than $3/q$. Set $t_0 = \lceil (q-1)\theta \rceil$. For each $0 \leq \ell \leq K-q$, the values with $x = \ell, \dots, \ell+q-1$ give mesh at most $3/q$ in

$$[\ell\theta + t_0, \ell\theta + K - 1].$$

Both endpoints are available by taking $x = \ell$. For finite-coefficient legality, write $x = \ell + j$. Subtracting $\ell\theta$, any retained phase point is $j\theta + y \in [t_0, K-1]$; $t_0 \geq j\theta$ gives $y \geq 0$, and the upper bound gives $y \leq K-1$. Thus no coefficient outside the finite square is used.

These windows overlap. Their union is

$$[s, t] = [t_0, \theta(K-q) + K - 1],$$

and $t - s > K + 1$, since $K \geq 8q$. In particular, for each $\xi \in [s, t]$ there are integers $0 \leq x, y < K$ with

$$\xi \leq \theta x + y < \xi + 3/q.$$

We now give a direct construction of legal integer representations. Suppose an old represented interval $[a, b] \subseteq P_n$ has width

$$W = b - a \geq 3\lambda/q + E_{n,k}.$$

Put $A = \lambda s + b - E_{n,k}$ and $B = \lambda t + b - E_{n,k}$. For any integer $z \in [A, B]$, apply the coefficient estimate to $\xi = (z - b + E_{n,k})/\lambda$. The corresponding actual suffix sum v satisfies

$$z - b \leq v < z - b + E_{n,k} + 3\lambda/q \leq z - a.$$

Thus $z - v \in [a, b]$ and $z = (z - v) + v$ is a legal representation. The old interval uses indices below n , and v uses only indices in $[n, n+k)$, so the supports are disjoint.

The represented integer interval has width

$$\lfloor B \rfloor - \lceil A \rceil > \lambda(t - s) - 2 > \lambda K,$$

where $\lambda \geq 2$ and $t - s > K + 1$. Its width exceeds the next unused weight $b_{n+k} = \lfloor \lambda K \rfloor$. Lemma 2.2 with unit gap then proves completeness.

Consequently, incompleteness forces $R_n < 3\lambda/q + E_{n,k}$. If $q \geq \lambda$, (9.1) and integrality give

$$R_n \leq 2(K_{n+k} - K_n) + 4.$$

Combining this with FE-R yields

$$K_{n+k} \geq K_n + \frac{c_0}{2} e^{aK_n} - 3. \quad (\text{DB})$$

This holds for every $n \geq 1$ and every reduced rational approximant with $q \geq 2^n \beta$ and $|\theta - p/q| < 1/q^2$.

10. Good rational approximants and long sparse windows

We call a reduced rational $r = p/q$, with $q > 0$, good if

$$|\theta - r| < q^{-2}.$$

Dirichlet approximation gives good rationals with arbitrarily large denominators when θ is irrational. For completeness, bounded denominators would allow only finitely many good rationals, whereas Dirichlet's estimates with increasing bounds force their errors to tend to zero. None can equal θ .

For an integer $b \geq 2$, let $D(b)$ be the least denominator at least b of a good rational. Dirichlet's theorem with bound $D(b) - 1$ supplies a reduced rational p/q such that

$$q < D(b), \quad |\theta - p/q| \leq \frac{1}{D(b)q}.$$

This rational is itself good, because $q < D(b)$. Minimality of $D(b)$ therefore gives $q < b$. We shall use both the good rational at the crossing denominator $D(b)$ and this pre-crossing rational; they serve different purposes.

10.1 Matching layers and large advances

Put

$$b(n) = \lceil 2^n \beta \rceil, \quad D_n^* = D(b(n)), \quad k(D) = \lceil \log_2(8D) \rceil, \quad f(n) = n + k(D_n^*).$$

Let $m(D) = \lfloor \log_2(D/\beta) \rfloor$ for $D \geq \beta$, and set

$$C_\beta = \lceil \log_2 \lceil 16\beta \rceil \rceil.$$

Then

$$2^{m(D)} \beta \leq D < 2^{m(D)+1} \beta, \quad k(D) \leq m(D) + C_\beta.$$

Apply DB to a good rational with denominator D_n^* . Under incompleteness,

$$K_{f(n)} \geq K_n + \frac{c_0}{2} e^{aK_n} - 3 \quad (n \geq 1).$$

We claim that $f(n) > n^3$ for arbitrarily large n . Otherwise, monotonicity of K_n , its divergence to infinity, and exponential growth would give

$$K_{n^3} \geq K_n^4$$

for all sufficiently large n . Choose such an $n_0 > 1$ with $K_{n_0} \geq 2$. Iteration yields

$$2^{4^r} \leq K_{n_0}^{4^r} \leq K_{n_0^{3^r}} \leq n_0^{3^r},$$

which is impossible as $r \rightarrow \infty$, since $(4/3)^r \rightarrow \infty$. This is the cubic-advance argument in the formalization.

At a matching layer $m = m(D) \geq 1$, the crude bound $E_{m,k(D)} \leq 2k(D)$ in Section 9, together with FE-R, gives

$$c_0 e^{aK_m} \leq 2m + 2C_\beta + 5. \quad (10.1)$$

Indeed, $R_m < 3 + 2k(D)$, so $R_m + 2 < 2k(D) + 5$. We apply this estimate only to matching layers.

10.2 Simultaneous window estimates

Choose an arbitrarily large n for which $f(n) > n^3$, put $D = D_n^*$ and $m = m(D)$, and set $T = m - 1$. For $n \geq C_\beta + 4$, the bound $k(D) \leq m + C_\beta$ implies $T \geq n^2$.

Take the pre-crossing rational $r = p/q$ constructed above, so that

$$q < b(n), \quad |\theta - r| \leq \frac{1}{Dq}.$$

Since $D \geq 2^n \beta$, these rationals approach θ as $n \rightarrow \infty$; in particular, $1 < r < 2$ for all sufficiently large choices. Define its binary height by

$$H = \lceil \log_2(p + q + 1) \rceil.$$

As $p < 2q$ and $q < \lceil 2^n \beta \rceil \leq 2^n \lceil \beta \rceil$, we have $H \leq n + C'_\beta$, where $C'_\beta = \lceil \log_2(3 \lceil \beta \rceil) \rceil$. Taking $n \geq C'_\beta$ gives

$$H^2 \leq 4T.$$

For $0 \leq i \leq T$, put $\delta_i = qa_i - pb_i$. The approximation implies

$$2^T |q\alpha - p\beta| \leq \frac{2^{m-1}\beta}{D} \leq \frac{1}{2}.$$

The floor errors lie in $[0, 1)$, so

$$|\delta_i| < p + q + 1 \leq 2^H \quad (0 \leq i \leq T).$$

Finally, (10.1) at the crossing denominator D and monotonicity of K give

$$c_0 e^{aK_T} \leq 2T + 2C_\beta + 7.$$

Thus there is a fixed constant $L > 0$ such that, on arbitrarily large windows, all the following hold simultaneously:

$$H^2 \leq 4T, \quad K_T \leq L \log T, \quad |\delta_i| < 2^H \quad (0 \leq i \leq T), \quad p/q \rightarrow \theta. \quad (10.2)$$

Here and below $\log$ denotes the natural logarithm. More precisely, for every prescribed error tolerance and lower bound on T , we can choose a window satisfying the displayed estimates and that tolerance. The good crossing rational supplies the event bound, while the pre-crossing rational supplies the small height and all-layer precision. No continued-fraction indexing is required.

11. The bounded-spacing contradiction (BG)

Assume the sequence is incomplete and, for some integer $R \geq 2$ and threshold n_0 , every $n \geq n_0$ has an event in $(n, Rn]$. Section 6 supplies this condition with $R = 4$. We derive a contradiction using the windows from Section 10.

11.1 Exact layers

Call i exact if $\delta_i = qa_i - pb_i = 0$. If the next H conversions after layer i are zero, then

$$\delta_{i+H} = 2^H \delta_i.$$

For $i + H \leq T$, the bound $|\delta_{i+H}| < 2^H$ and integrality force $\delta_i = 0$. Hence every nonexact layer $i \leq T - H$ sees an event among the next H arrivals. Each event can be counted by at most H such layers. Including the last H layers separately, we obtain

$$\#\{0 \leq i \leq T : \delta_i \neq 0\} \leq HK_T + H =: E. \quad (11.1)$$

By (10.2), $E = O(\sqrt{T} \log T)$.

11.2 A uniform lower bound on nontrivial return cost

Fix a positive integer k . Let

$$\mathcal{A} = \{0\} \cup \{2^{-j} : j \in \mathbb{N}\}, \quad \mathcal{S}_k = \underbrace{\mathcal{A} + \cdots + \mathcal{A}}_{k \text{ times}}.$$

The set $\mathcal{A}$ is compact, and every element is rational. Therefore $\mathcal{S}_k$ is compact and consists of rational numbers. The set of ratios

$$\mathcal{C}_k = \{x/y : x, y \in \mathcal{S}_k, y \geq 1/2\}$$

is also compact and consists entirely of rational numbers. A fixed irrational θ consequently has a neighbourhood disjoint from $\mathcal{C}_k$.

Suppose $a < b$ are exact layers and the interval of arrivals $(a, b]$ contains at least one event. Unrolling the floor recurrences gives nonnegative binary words U, V with

$$a_b = 2^{b-a}a_a + U, \quad b_b = 2^{b-a}b_a + V.$$

Exactness at both ends implies $qU = pV$. Since some event occurs, the words are not both zero; since $p, q > 0$, both are positive and $U/V = p/q$.

If $(a, b]$ contained at most k events, divide U and V by the largest power of two appearing in either word. Each normalized word belongs to $\mathcal{S}_k$, and at least one is at least one. The ratio lies in $(1, 2)$, so the normalized denominator is at least $1/2$. Hence $p/q \in \mathcal{C}_k$.

It follows that, once p/q is sufficiently close to θ , every nontrivial return between exact layers costs more than k events. This estimate is uniform over the common multiplier in $U = cp, V = cq$; we impose no bound on that multiplier.

11.3 Geometric capacity

Put $B = 2R + 1$ and choose a fixed positive integer k with $k \geq 8L \log B$. Choose a window from Section 10 sufficiently far out that the return-cost conclusion for this k holds. Write $K = K_T$ and set

$$x = \max(n_0 + 1, E + 1), \quad r = \lfloor K/k \rfloor + 1.$$

Since $E = O(\sqrt{T} \log T)$, we have $x \leq T^{3/4}$ for all sufficiently large windows. Also, $K \leq L \log T$ implies

$$B^r \leq B T^{1/8}.$$

Thus, after enlarging the lower bound on T ,

$$2B^r x \leq 2B T^{7/8} \leq T.$$

For each $i = 0, \dots, r$, the integer interval $[B^i x, 2B^i x]$ lies in $[0, T]$ and contains more than E layers. By (11.1), it contains an exact layer z_i . Consecutive choices satisfy

$$z_{i+1} \geq B^{i+1} x > R(2B^i x) \geq R z_i.$$

Moreover, $z_i \geq n_0$, so there is an event in $(z_i, R z_i]$ and hence in $(z_i, z_{i+1}]$. The r return intervals are disjoint. Each costs more than k events, and in particular at least k . Therefore $r k \leq K_T = K$, contradicting $r = \lfloor K/k \rfloor + 1$.

This proves that irrationality and bounded event spacing are incompatible with incompleteness. Sections 6 and 7 now establish the theorem. $\square$

12. Correspondence with the Lean proof

The formalization uses integer-valued floor sequences and finite sets of natural-number indices. The final statements in `Statements.lean` are fixed independently of the intermediate proof modules.

Argument in this exposition	Principal Lean modules
Normalization and distinct retained values, §§1 and 7	Normalization, SetBridge, Main
Finite cone certificate, §3 and Appendix A	Basic, CertificateData, Certificate
Mesh lemmas and permanent descent, §§2–6	CyclicGaps, Mesh, InitialMesh, PermanentMesh, FloorDescent, EventGaps
Finite-event decay and interval seeds, §8	FECounting, FEMissingRuns, FEShift, PeriodChange, EventBoundary, FERecurrence, EventDecay, FE, FER
Legal coefficient coverage and digit budget, §9	DBPhases, DBWindows, DBDigits, DBCover, DB
Good rational crossings and sparse windows, §10	RationalWindows, CubicGrowth, DBScale, BGWindowBounds, BGWindows
Exact layers, return cost, and contradiction, §11	BGExactLayers, BGSparseCompact, BGBinaryRatio, BGReturns, BGGeometric, BGCapacity, BG

The `detailed theorem map` identifies the principal declarations. The two final results are `Dyadic354.erdos354_part_i` and `Dyadic354.erdos354_strong_completeness`. The extracted-definition adapter also proves the positive part-(i) target in the exact upstream definitions.

The project fixes Lean 4.27.0 and its dependency revisions. Our build and audit check all 381 local theorems and require every transitive axiom set to be a subset of `{propext, Classical.choice, Quot.sound}`. The certificate computation is checked by the kernel. The `verification records` give the source revisions, commands, and actual build output.

References

1. T. F. Bloom, [Erdős Problem 354](#): problem statement and references to Graham and Hegyvári.
2. The Mathlib Community, [Diophantine approximation](#), pinned revision `a3a10db0`: the Dirichlet approximation results used in the good-rational construction.
3. The Formal Conjectures Authors, [Erdős Problem 354](#), pinned revision `a748dd91`: provenance of the formal target definitions.

Appendix A. Complete finite mask certificate

The bit order for the first four weights is a_1, b_1, a_2, b_2 , starting at the least significant mask bit. A triple (M_0, M_1, J) specifies the two first-four masks and the same third-pair subset added to both. $J=0, 1, 2, 3$ denotes neither, a_3, b_3 , or both. All intervals are reconstructed as in Section 3.

First digits	Second digits	Complete mask chain
(1, 0)	(0, 0)	(3, 4, 0), (6, 5, 0), (2, 1, 2), (3, 4, 2) (6, 5, 2), (14, 7, 2), (2, 1, 3), (3, 4, 3) (6, 5, 3)
(1, 0)	(0, 1)	(10, 9, 0), (11, 12, 0), (7, 13, 0), (2, 1, 1) (3, 4, 1), (6, 5, 1), (2, 1, 3), (3, 4, 3) (6, 5, 3)
(1, 0)	(1, 0)	(10, 3, 0), (6, 5, 0), (2, 1, 2), (10, 3, 2) (6, 5, 2), (14, 7, 2), (2, 1, 3), (10, 3, 3) (6, 5, 3)
(1, 0)	(1, 1)	(3, 9, 0), (6, 5, 0), (2, 1, 2), (7, 13, 0) (2, 1, 1), (6, 5, 2), (3, 9, 1), (6, 5, 1) (2, 1, 3), (14, 13, 1), (3, 9, 3), (6, 5, 3)
(0, 1)	(0, 0)	(9, 10, 0), (12, 11, 0), (1, 3, 2), (9, 10, 2) (12, 11, 2), (7, 13, 2), (1, 2, 3), (4, 3, 3) (5, 6, 3)
(0, 1)	(0, 1)	(9, 10, 0), (12, 11, 0), (1, 3, 2), (9, 10, 2) (12, 11, 2), (13, 14, 2), (1, 2, 3), (4, 3, 3) (5, 6, 3)
(0, 1)	(1, 0)	(3, 9, 0), (5, 6, 0), (1, 2, 2), (1, 3, 2) (3, 9, 2), (5, 6, 2), (3, 9, 1), (5, 6, 1) (1, 2, 3), (1, 3, 3), (3, 9, 3), (5, 6, 3)
(0, 1)	(1, 1)	(8, 10, 0), (6, 9, 0), (1, 2, 2), (8, 10, 2) (4, 6, 2), (8, 10, 1), (6, 9, 1), (1, 2, 3) (1, 3, 3), (9, 10, 3), (5, 6, 3)
(1, 1)	(0, 0)	(3, 9, 0), (6, 11, 0), (0, 1, 2), (1, 8, 2) (3, 9, 2), (6, 11, 2), (7, 14, 2), (0, 1, 3) (1, 8, 3), (3, 9, 3), (6, 11, 3)
(1, 1)	(0, 1)	(3, 8, 0), (6, 9, 0), (7, 11, 0), (1, 3, 2) (13, 15, 0), (3, 8, 1), (6, 9, 1), (7, 11, 1) (3, 8, 3), (6, 9, 3), (5, 7, 3)
(1, 1)	(1, 0)	(3, 4, 0), (4, 5, 0), (5, 12, 0), (7, 13, 0) (0, 1, 1), (1, 8, 1), (3, 4, 1), (4, 5, 1) (5, 12, 1), (7, 13, 1), (3, 4, 3), (4, 5, 3) (5, 12, 3)
(1, 1)	(1, 1)	(3, 4, 0), (6, 11, 0), (7, 12, 0), (3, 4, 2) (6, 11, 2), (3, 4, 1), (6, 11, 1), (7, 12, 1) (3, 4, 3), (6, 11, 3)